

Boron–Zinc Exchange in The Diastereoselective Arylation of Sugar-Based Aldehydes: Stereoselective Synthesis of (+)-7-*epi*-Goniofufurone and Analogues

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Received: 12.04.2013; Accepted after revision: 15.05.2013

Abstract: The substrate-controlled diastereoselective arylation of chiral aldehydes readily available from carbohydrates is described, using the boron–zinc exchange reaction to generate the transferable aryl groups. The methodology developed was applied to the total synthesis of the styryllactone (+)-7-*epi*-goniofufurone and analogues thereof.

Key words: arylzinc reagents, boron–zinc exchange, diastereoselective addition, chelation control, styryllactones

Introduction

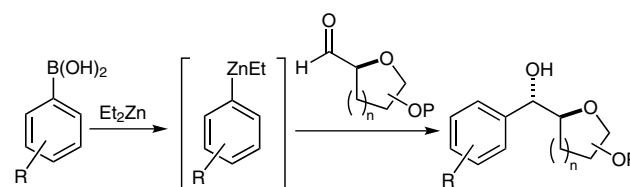
The stereoselective addition of nucleophiles to carbonyl compounds is a fundamental transformation in organic synthesis.¹ For a successful synthesis strategy toward target molecules that require selectivity in the addition of an organometallic reagent to an aldehyde or ketone, the choice of the organometallic reagent to be used is of crucial importance. Organozinc reagents have played an important role, especially due to their low reactivity, which makes them suitable for the use in asymmetric catalysis. Indeed, diethylzinc addition has become a benchmark reaction for the evaluation of the efficiency of a given chiral catalyst for the alkylation of aldehydes.² Stereoselective arylation, however, is a more challenging task due to the fact that arylzinc reagents are far more reactive than dialkylzinc reagents and they are transferred without the need of a ligand to assist in the reaction. Actually, several research groups have been involved in solving this problem and a promising strategy is to generate mixed arylalkylzinc reagents that are considerably less reactive than the diarylzinc reagents.^{3,4} In this context, one of the most appealing methods for the generation of such species is by the boron-to-zinc exchange reaction between arylboronic acids and diethylzinc.^{5,6} This reaction has many advantages, such as the many arylboronic acids that are commercially available or easily prepared, the smooth B–Zn exchange to generate the reactive ArZnEt, and the high preference of aryl versus alkyl transfer.⁷

Despite these advances in the generation of efficient arylzinc nucleophiles and their successful application in the

catalytic asymmetric addition to aryl aldehydes, their use in organic synthesis is still underexplored. Besides the above-mentioned reactions, arylalkylzinc reagents have been used in asymmetric nickel-catalyzed Negishi cross-couplings,⁸ and in iron-catalyzed⁹ and catalyst-free¹⁰ arylations of benzyl halides.

Carbohydrates have always been among the most outstanding classes of biomolecules and they have long been used as building blocks for the synthesis of biologically active natural products. The popularity of using sugars as starting material for syntheses is due to their low cost, abundance, and, most importantly, their multitude of functional, stereochemical, and conformational features that allows advantage to be taken of their inherent molecular characteristics and turn them into interesting synthetic intermediates.¹¹

Recently, our research program focused in the use of sugars as templates for the synthesis of chiral molecules,¹² and connected with our interest in the interesting reactivity of arylalkylzinc reagents,¹³ we have investigated the stereoselectivity of the arylation of enantiopure furanose-aldehydes derived from carbohydrates (Scheme 1).¹⁴



Scheme 1 General approach for the diastereoselective arylation of sugar-derived aldehydes

Styryllactone natural products were envisaged as a target family of relevance that could profit from such designed reactions. The styryllactones are isolated from different sources and, depending on their structure, have been described to possess important biological activity such as anti-inflammatory, antibiotic, antifertility, immunosuppressant, and anticancer properties.¹⁵ Among the different structural features displayed by these compounds, many of them have in common a stereodefined benzylic alcohol/ether moiety (Figure 1).

We wish to report herein the full account on the substrate-controlled diastereoselective arylation of chiral aldehydes

Biographical Sketches



Ana Dionéia Wouters was born in Horizontina, RS, Brazil in 1983. She received her B.Sc. in chemistry in 2006 from the Universidade Federal de Santa Maria. In 2008, she completed her M.Sc. degree at the same university under the supervi-

sion of Prof. Nilo Zanatta. She received her Ph.D. degree from the Universidade de São Paulo in 2013 under the direction of Prof. Diogo S. Lüttke; she worked with the synthesis of sugar derivatives and its applications in enantioselective and

diastereoselective processes. During her Ph.D. studies she was a visiting scientist at the Leibniz Institute of Plant Biochemistry with Prof. Ludger A. Wessjohann in Halle, Germany.



Adrieli Barros Bessa was born in Sao Paulo, SP, Brazil in 1988. In 2009, she entered the research group of Prof. Diogo S. Lüttke as an undergraduate student at the University of São Paulo; her research focused on the design and preparation of new chiral ligands and catalysts from readily available sources, such as amino acids, and their application in asym-

metric catalysis. In 2011, she entered the clinical research staff in kidney transplants at the Hospital do Rim e Hipertensão, where she currently holds the position of Study Coordinator. This team conducts multicenter studies researching new treatments for kidney transplant recipients. She received her B.S. in pharmacy in 2012 from the Faculdades

Oswaldo Cruz. In 2013, she entered the graduate program at the Federal University of Sao Paulo, where her studies for a master's degree, under the supervision of Prof. Jose Osmar Medina Pestana are ongoing; she is researching the impact of pharmaceutical intervention on the intervariability of blood concentrations of tacrolimus.



Maraisa Sachini was born in 1984, and grew up in Veranópolis-RS, Brazil. She obtained her B.S. in chemistry in 2008 and M.Sc. in 2011 at the Federal University of Pelotas (UFPEl) under the supervision

of Prof. Gelson Perin on the synthesis of vinyl chalcogenides in a green chemistry context. In 2011, she entered the graduate program at the Federal University of Rio Grande do Sul (UFRGS). Her Ph.D. is

currently in progress, developing research on carbohydrates as raw material in the synthesis of chiral molecules, under the supervision of Prof. Diogo S. Lüttke.



Professor Wessjohann studied chemistry in Hamburg (Germany), Southampton (UK), and Oslo (Norway). He earned his doctorate in 1990 with Prof. de Meijere in Hamburg. After a short period as a lecturer in Brazil, he became a postdoctoral Feodor-Lynen fellow of the Alexander von Humboldt foundation with Prof. Paul Wender at Stanford (USA) working on the total synthesis of Taxol®. After an assistant professorship in Munich (LMU, 1992–1998) he

was appointed to the Chair of Bioorganic Chemistry at the Vrije Universiteit Amsterdam (NL), working on organometallic chemistry and biocatalysis. Since 2001 he is the Director of the Department of Bioorganic Chemistry at the Leibniz Institute of Plant Biochemistry (IPB) in Halle (Germany), and in parallel holds the Chair of Natural Product Chemistry at the Martin Luther-University Halle-Wittenberg. Since 2010 he is the Managing Director of IPB.

Prof. Wessjohann focuses on the discovery, synthesis, and application of natural products and derivatives. He has over 250 publications, over 20 patent applications, and is co-founder of four companies. He is member of many boards and commissions, and he has received several scholarships, prizes, and honors, e.g. the Microsoft IT Founders Award, or the honorary membership of the Argentinean Society of Synthetic Organic Chemistry.



Diogo S. Lüttke (born in 1979) grew up in Paraíso do Sul, a small town in Southern Brazil. He obtained his degree in chemistry from the Federal University of Santa Maria (UFSM) in 2000. In 2001, he entered the graduate program at the same University, where in 2005 he completed his Ph.D. under the supervision of Prof.

Antonio L. Braga, working on the synthesis of chiral selenium-containing amino acid derivatives and peptides as well as with catalytic enantioselective synthesis mediated by chiral selenium ligands. After a postdoctoral stint in the group of Prof. Ronaldo A. Pilli, at the State University of Campinas (UNICAMP), he started his in-

dependent career in 2007, at the University of São Paulo (USP). Since 2011 he has been at the Institute of Chemistry of the Federal University of Rio Grande do Sul (UFRGS). He is the recipient of the CNPq Research Productivity Fellowship (level 2) and is an Affiliated Member of the Brazilian Academy of Sciences.

and the application of the methodology developed in the total synthesis of (+)-7-*epi*-goniofufurone and analogues thereof.

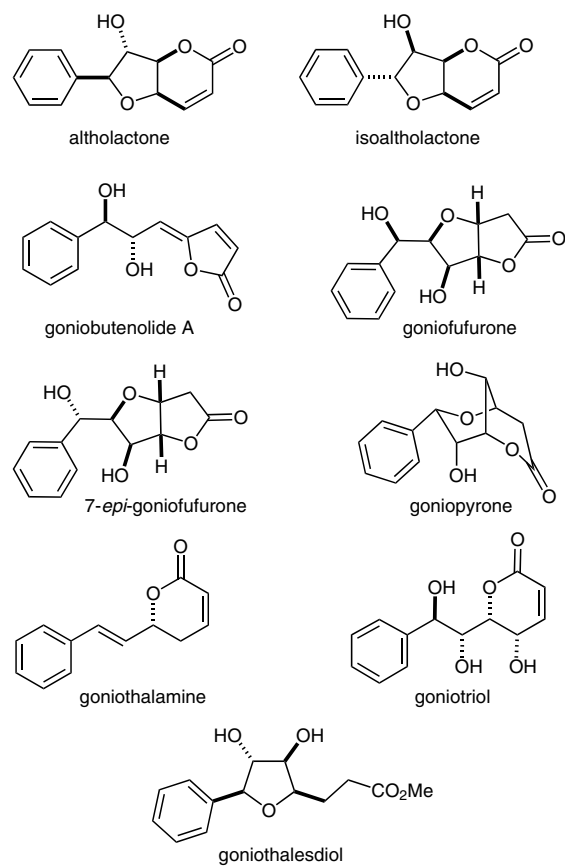


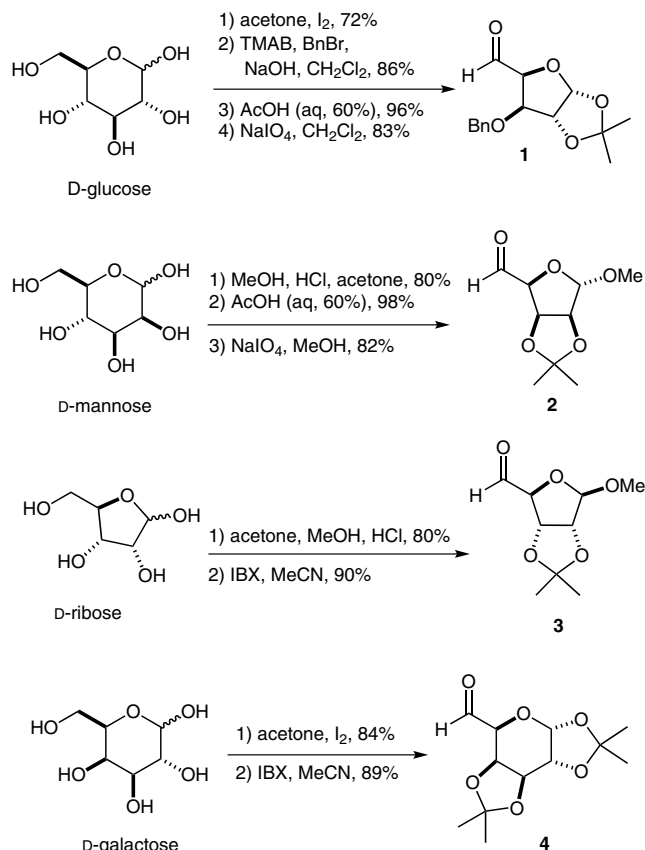
Figure 1 Selected members of the styryllactone family of natural products

Results and Discussion

The sugar-derived aldehydes required for our study were prepared starting from the parent carbohydrates in short synthetic sequences (2 to 4 steps) as depicted in Scheme 2.^{16–19}

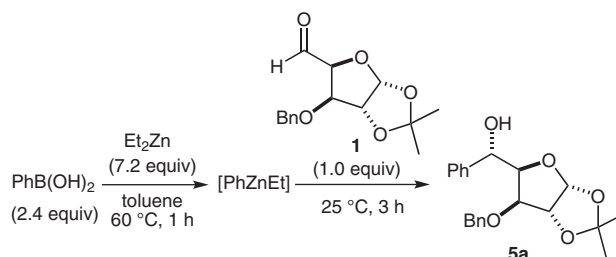
We chose as starting materials the carbohydrates D-glucose, D-mannose, D-ribose, and D-galactose in order to obtain aldehydes **1–4** with varied substitution patterns and, more importantly, with different stereochemistry.

The initial studies for the arylation reaction focused on optimization of the reaction conditions using phenylboronic acid and aldehyde **1**, derived from D-glucose. The arylzinc reagents were prepared in situ by a boron–zinc exchange using arylboronic acids (2.4 equiv) and diethylzinc (7.2 equiv). After screening of solvents such as toluene, dichloromethane, tetrahydrofuran, and hexane, we found that the best yields and diastereoselectivities were achieved in toluene. Further optimization of temperature and reaction time led us to our best conditions for the aryl-transfer reaction (toluene, 25 °C, 3 h), in which the desired product **5a** was obtained in 73% yield and an excel-



Scheme 2 Synthesis of the aldehydes

lent diastereomeric ratio of >20:1 (Scheme 3). Attempts to improve the yield running the reaction at a higher temperature (60 °C or 100 °C) led to an erosion of the diastereoselectivity of the reaction.

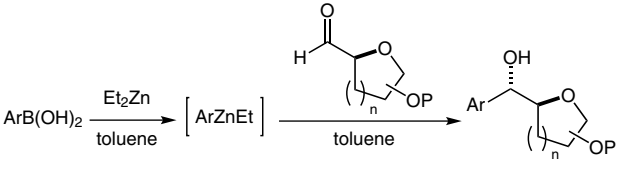
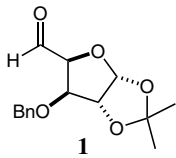
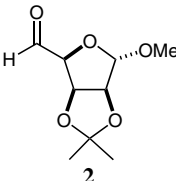
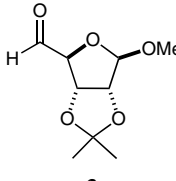
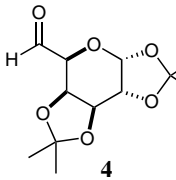


Scheme 3 Phenyl-transfer reaction to aldehyde **1**

The absolute stereochemistry of the newly formed stereocenter was assigned by comparison of the analytical data of **5a** with literature data²⁰ and, after completion of its total synthesis, of the natural product (+)-7-*epi*-goniofufurone (see Scheme 7 for further discussion). The use of the boron–zinc exchange to generate the actual transferable phenyl group represents an important advance over previously described reaction conditions using more conventional organometallics.^{20,21} For example, phenylmagnesium bromide addition to aldehyde **1** resulted in moderate to good diastereomeric ratios (1.6 to 14:1). Even more important is the fact that the present methodology is not lim-

ited to phenyl addition and allows the transfer of a wide range of aryl groups, which is a desirable feature for the synthesis of natural product analogues for biological activity screenings. Therefore, after assessing the best reaction conditions, we took advantage of variety of commercially available arylboronic acids to further expanded our studies to a broader range of aryl groups. Along these lines, we studied the arylation of aldehyde **1** with different arylzinc reagents (Table 1, entries 1–7) and we found that both electron-rich and electron-poor ones are transferred, resulting in the desired products in good yields and in very high diastereomeric ratios. The reaction proceeded cleanly and with high conversion of the aldehyde as observed by TLC analysis. In some cases isolated yields were lower than conversions due to the high water solubility of the products.

Table 1 Diastereoselective Addition of Arylzinc Reagents to Sugar-Based Aldehydes

					
Entry	Aldehyde	Ar	Product	Yield ^a (%)	dr ^b
1		Ph	5a	73	>20:1
2		3,4,5-F ₃ C ₆ H ₂	5b	66	>20:1
3		4-ClC ₆ H ₄	5c	72	>20:1
4		4-BrC ₆ H ₄	5d	73	14:1
5		2-MeC ₆ H ₄	5e	70	16:1
6		4-MeOC ₆ H ₄	5f	75	>20:1
7		4-MeC ₆ H ₄	5g	68	>20:1
8		Ph	6a	78	5:1
9		4-ClC ₆ H ₄	6b	75	7:1
10		4-MeOC ₆ H ₄	6c	74	4:1
11		2-MeC ₆ H ₄	6d	78	>20:1
12		4-MeC ₆ H ₄	6e	79	3:1
13		Ph	7a	78	>20:1
14		4-ClC ₆ H ₄	7b	77	>20:1
15		2-MeC ₆ H ₄	7c	76	>20:1
16		4-MeOC ₆ H ₄	7d	77	>20:1
17		4-BrC ₆ H ₄	7e	72	>20:1
18		Ph	8a	74	2.7:1
19		4-MeC ₆ H ₄	8b	72	2.7:1
20		4-MeOC ₆ H ₄	8c	65	2.5:1

^a Isolated yields.

^b Determined by ¹H NMR of the crude product.

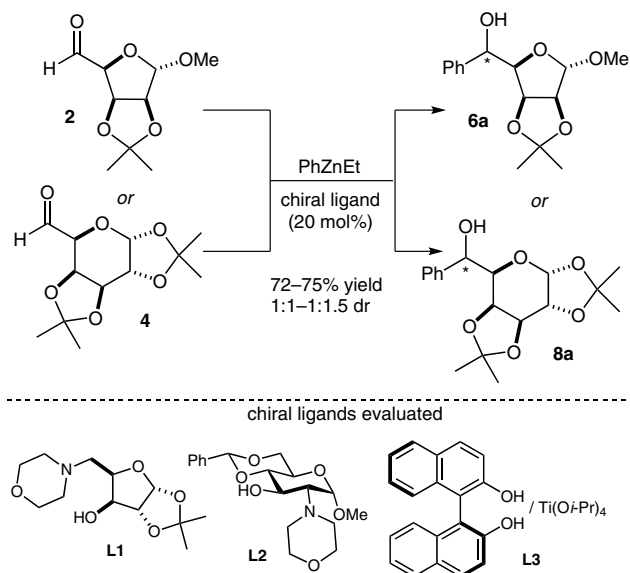
Encouraged by the success achieved in the arylation of aldehyde **1**, which allowed us to introduce different aryl moieties within the carbohydrate-derived skeleton, we expanded the studies to the furanoside aldehydes **2** and **3** and to pyranoside aldehyde **4**.

We first examined the behavior of aldehyde **2**, derived from D-mannose and we found that performing the arylation at room temperature took six hours for full conversion of the starting materials. After heating the mixture to 60 °C, we could reduce the reaction time to three hours with improved yields and the same diastereoselectivity. Under these conditions, good yields of the arylation products **6a–e** were obtained, however with moderate diastereomeric ratios (Table 1, entries 8–12). An interesting result was obtained with the more hindered 2-tolyl derivative, that gave the corresponding product **6d** in 78% yield and an excellent >20:1 dr (entry 11). Conversely, when we tested the ribose-derived aldehyde **3**, perfect stereoselectivities of the arylation reactions were found (entries 13–15). In all cases studied the desired products **7a–e** were isolated in high yields and excellent diastereoselectivities of >20:1, irrespective of the nature of the arylzinc nucleophile. The behavior of the arylation reaction is the same with arylzinc substituted with electron-donating, electron-withdrawing or bulky groups. Again, the best results were obtained by performing the reaction at 60 °C. Monitoring the consumption of the aldehyde by TLC revealed that the reaction was complete after one hour.

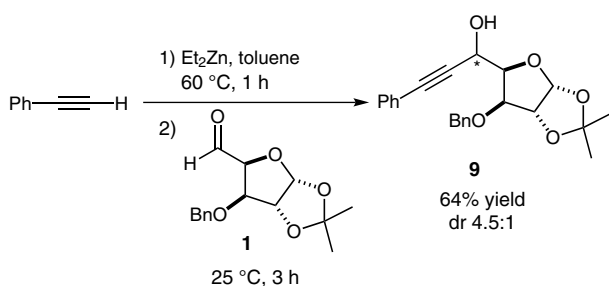
In contrast to the previous success with the ribose-derived aldehyde, the arylation reaction of the pyranoside aldehyde **4**, derived from D-galactose led to decreased diastereoselectivities (entries 18–20). For the three cases studied, the corresponding benzylic alcohols **8a–c** were isolated in good yields and low diastereomeric ratios of 2.5:1 to 2.7:1 only.

In an attempt to improve the diastereomeric ratios observed for aldehydes **2** and **4**, we tested the use of a chiral ligand as an additive to verify if we could override or improve the selectivity issues imposed by the substrate (Scheme 4). Unfortunately, we only observed an erosion of the diastereoselectivity and the products **6a** and **8a** were obtained in 72–75% yield. The chiral ligands used as additives led to an unselective addition or to a slight inversion of the diastereomeric ratio previously observed in the absence of the additive, indicating that there is a mismatched relationship that led to a partial, but not significant, override of the substrates stereoreinduction.

In another reaction we studied the alkynylation of aldehyde **1** using an alkynylzinc reagent generated in situ by the deprotonation of phenylacetylene with diethylzinc. The resulting alkynylalkylzinc reagent reacted with aldehyde **1** and the product was formed in 64% yield and only with low diastereomeric ratios of 4.5:1 (Scheme 5). Despite several attempts to improve the observed stereoselectivity by changing reaction times, temperatures, and solvents, we were not able at this point to achieve high levels of diastereoselectivity.



Scheme 4 Evaluation of the effect of chiral ligands as additives



Scheme 5 Alkynylzinc addition to aldehyde 1

In order to understand the mechanism of the diastereoselective addition of the arylzinc reagents to the chiral sugar-based aldehydes the pre-transition-state assemblies depicted in Scheme 6 are proposed.²² We believe that the reaction is a chelation-controlled process and coordination of zinc to the carbonyl group and to an oxygen atom of the core structure of the aldehydes may be critical for the reaction outcome. Therefore, the Lewis acidic ArZnEt would coordinate to the aldehyde, which might act as a bidentate Lewis base. This coordination mode would make the aldehyde behave similarly to amino alcohols in standard diorganozinc additions, enhancing the electrophilicity of the carbonyl group.²

The selectivity of the reaction would thus be the result of an attack of a second arylzinc nucleophile on the energetically most favored chelate conformation. In the case of aldehyde 1, the most favored conformation **A** results from chelation of the zinc atom to the carbonyl and the endocyclic furanoside oxygen. Attack of the nucleophile would occur preferentially at the least hindered *si* face, delivering the corresponding product **5** as the major diastereomer. This proposed mechanism explains the stereochemical results observed with the new stereocenter

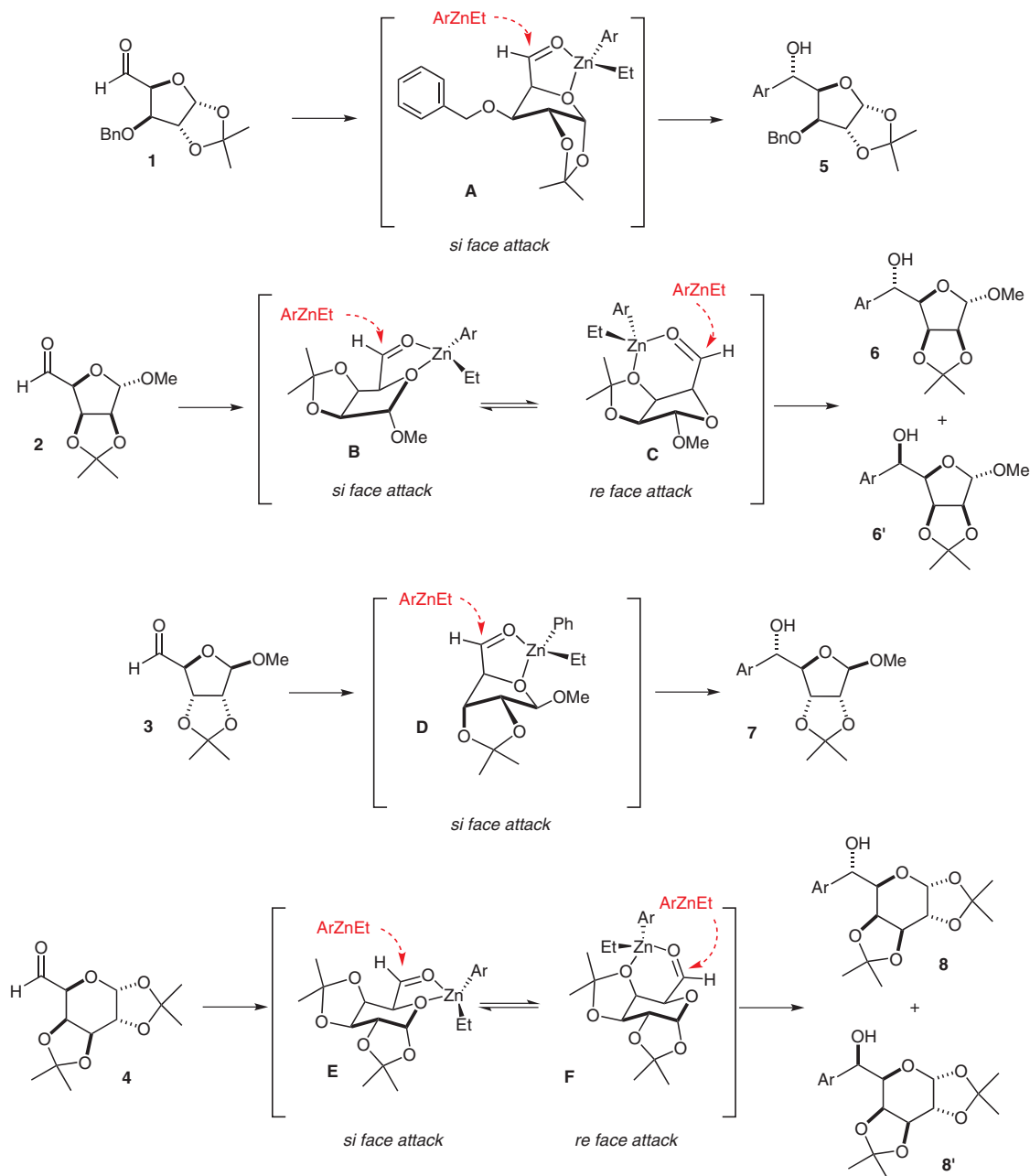
assigned by correlation with literature NMR data.²⁰ We believe that the other aldehydes **2–4** studied undergo reactions via similar chelation-controlled transition states. Unfortunately until now, we were unable to unequivocally assign the absolute stereochemistry of the new stereocenter in products **6–8**. All attempts to achieve suitable crystals for X-ray diffraction analysis failed. Despite this drawback, we can rationalize our results on the basis of what was observed for aldehyde 1. Accordingly, similar behavior can be proposed for the reaction with aldehyde 3. The pre-transition state assembly **D** with chelation between the aldehyde and the pyranoside oxygen would expose the *si* face to nucleophilic attack, leading to chiral alcohol **7a** as the major product.

On the other hand, for aldehydes **2** and **4**, the diastereoselectivity of the arylation reaction was lower than for aldehydes **1** and **3**. In the case of the mannose-derived aldehyde **2**, the opposite stereochemistry at the isopropylidene acetal moiety, compared to the ribose-aldehyde **3** can lead to two competing pre-transition states assemblies **C** and **D**. One possible pathway follows the rationalization given before and a 1,2-chelation involving the furanoside oxygen would lead to reaction at the *si* face. An alternative pathway can be envisioned where the *cis* relationship between the acetonide protecting group and the aldehyde would allow a different 1,3-coordination, leading to sterically dense **C**, in which the *re* face of the aldehyde would be more accessible to nucleophilic attack. The two competing transition states **B** and **C** would respectively lead to diastereomeric products **6** and **6'**. Similar trends can be found for galactose-derived aldehyde **4**, in which a *cis* relationship between the acetonide protecting group and the aldehyde can form structure **F** that can compete with the chelation with the pyranoside oxygen. Once again, these two competing 1,2- and 1,3-chelation pathways would lead to diastereomers **8** and **8'**.

Synthesis of (+)-7-*epi*-Goniofufurone and Analogues

The utility of the stereoselective arylation of aldehyde **1** leading to the chiral alcohol **5** can be exemplified with the synthesis of (+)-7-*epi*-goniofufurone, a natural product isolated from the bark of *Goniothalamus giganteus*, that has shown significant activity against human tumor cells (Scheme 7).²³ Actually, the synthesis of **5a** ($\text{R} = \text{H}$) itself constitutes a formal synthesis of the natural product, since it intercepts the synthesis reported by Popsavin and co-workers.^{21a} Using Popsavin's protocol, we could complete the total synthesis of (+)-7-*epi*-goniofufurone and of four analogues thereof with different substituents at the aryl moiety.²⁴

The synthesis was accomplished starting with an acidic hydrolysis of the acetonide protecting group in aqueous acetic acid. The corresponding triols **10a–e** reacted with Meldrum's acid in the presence of triethylamine to forge the required fused lactone rings in **11a–e**. Finally, hydro-



Scheme 6 Proposed pathways for the diastereoselective arylation of sugar-based aldehydes

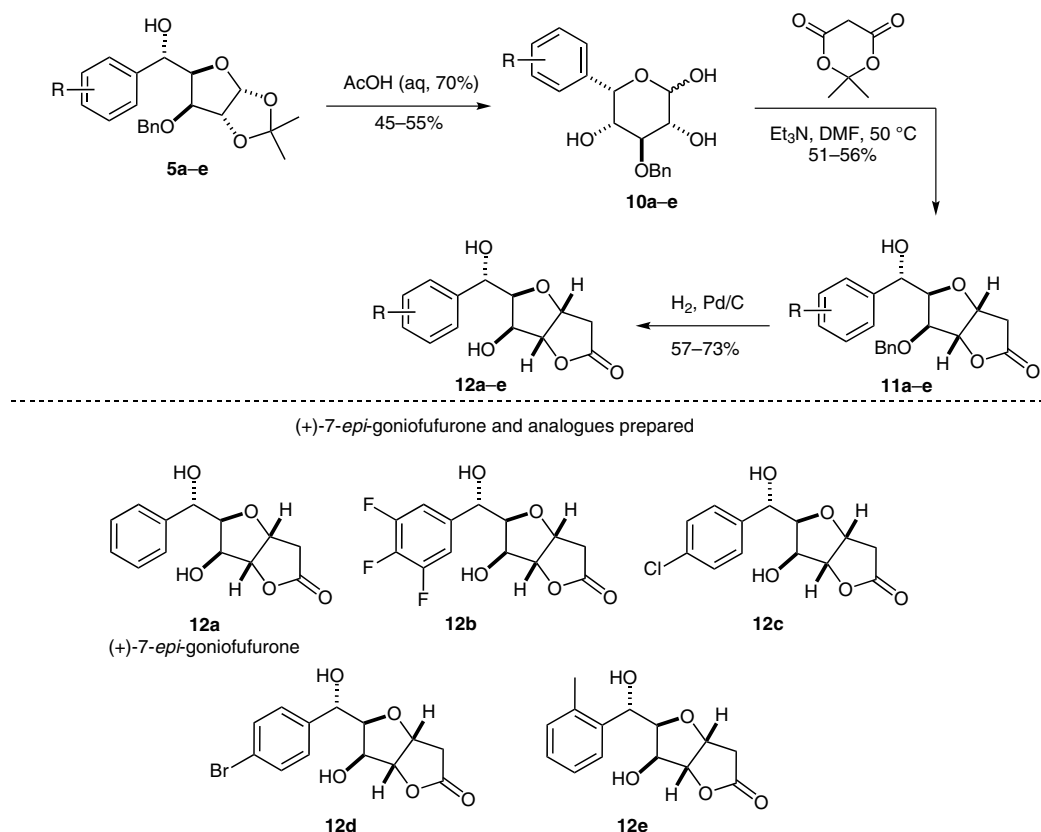
genolysis in the presence of palladium on carbon catalyst removed the benzyl group and delivered the desired products **12a–e** in good overall yields.

Importantly, the completion of the synthesis of (+)-7-epigoniofufurone (**12a**) corroborates the assignment of the stereocenter at the benzylic position created by arylzinc addition to aldehyde **1**.

Conclusion

In summary, we have reported the diastereoselective arylation reaction of chiral aldehydes, readily available from carbohydrates. Key to our strategy was the use of the

boron-to-zinc exchange reaction between arylboronic acids and diethylzinc to generate the reactive arylethylzinc nucleophiles. Excellent results were achieved with aldehydes **1** and **3**, derived from D-glucose and D-ribose, respectively. In contrast, the reaction with aldehydes **2** and **4**, obtained from D-mannose and D-galactose resulted in only moderate diastereoselection. The observed selectivities allowed us to propose chelation-controlled transition states in which the stereochemistry of the sugar oxygens relative to the aldehyde moiety have a key role for the level and orientation of diastereoselectivity in the reaction. Important to note is that our method leads to higher diastereoselectivity and structural variation when compared with the more traditional Grignard additions.



Scheme 7 Total synthesis of (+)-7-*epi*-goniofufurone and analogues

As an application, we completed the total synthesis of the natural styryllactone (+)-7-*epi*-goniofufurone and four additional analogues with different substituents at the aryl moiety.

^1H NMR were obtained at 300 MHz. Spectra were recorded in CDCl_3 solns, referenced to the solvent peak of residual CHCl_3 or TMS as reference. ^{13}C NMR were obtained at 75 MHz. Spectra were recorded in CDCl_3 solns, referenced to the solvent peak CDCl_3 . Optical rotations were obtained on a Perkin Elmer 341 Polarimeter. HRMS were obtained on a BrukerBioApex 70e. Column chromatography was performed using silica gel (230–400 mesh) following the methods described by Still.²⁵ TLC was performed using silica gel GF₂₅₄, 0.25-mm thickness. For visualization, TLC plates were either placed under UV light, or phosphomolybdic acid, followed by heating. Air- and moisture-sensitive reactions were conducted in flame-dried or oven-dried glassware equipped with tightly fitted rubber septa and under a positive atmosphere of anhyd argon. Reagents and solvents were handled using standard syringe techniques. Temperatures above r.t. were maintained by use of a mineral oil bath heated on a hotplate. Aldehydes **1–4** were prepared from D-glucose, D-mannose, D-ribose, and D-galactose, respectively, according to literature procedures.^{16–19}

Aryl-Transfer Reaction to Aldehyde **1**; General Procedure

In an atmosphere of argon, 1.5 M Et_2Zn in toluene (2.33 mL, 3.6 mmol, 7.2 equiv) was slowly added to a soln of arylboronic acid (1.2 mmol, 2.4 equiv) in anhyd toluene (2 mL). The mixture was stirred at 60 °C for 1 h and after this period, cooled to r.t. and a soln of the aldehyde (0.139 g, 0.5 mmol) was added. The mixture was stirred at r.t. for 3 h and then extracted with EtOAc (2 × 20 mL). The combined extracts were washed with brine and dried (MgSO_4), and

the solvent was evaporated using a rotary evaporator. The residue was purified by flash chromatography (hexane–EtOAc, 7:3).

3-*O*-Benzyl-1,2-*O*-isopropylidene-5-*C*-phenyl- β -L-ido-pentofuranose (**5a**)

Yellow oil; yield: 0.130 g (0.37 mmol, 73%).

$[\alpha]_{\text{D}}^{20}$ –28 (*c* 1, CHCl_3).

^1H NMR (300 MHz, CDCl_3): δ = 1.30 (s, 3 H), 1.48 (s, 3 H), 2.85 (br s, 1 H), 3.63 (d, J = 3.2 Hz, 1 H), 4.29–4.34 (m, 2 H), 4.55 (d, J = 11.5 Hz, 1 H), 4.59 (d, J = 11.5 Hz, 1 H), 5.05 (d, J = 7.4 Hz, 1 H), 6.02 (d, J = 3.7 Hz, 1 H), 7.28–7.38 (m, 10 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 26.1, 26.7, 71.7, 72.3, 82.0, 82.1, 84.4, 106.0, 111.8, 127.0, 127.5, 127.9, 128.2, 128.2, 128.4, 136.9, 139.0.

HRMS: m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{21}\text{H}_{24}\text{O}_5\text{Na}$: 379.1516; found: 379.1520.

3-*O*-Benzyl-5-*C*-3,4,5-fluorophenyl-1,2-*O*-isopropylidene- β -L-ido-pentofuranose (**5b**)

Yellow oil; yield: 0.136 g (0.33 mmol, 66%).

$[\alpha]_{\text{D}}^{20}$ –32 (*c* 1, CHCl_3).

^1H NMR (300 MHz, CDCl_3): δ = 1.32 (s, 3 H), 1.47 (s, 3 H), 3.21 (d, J = 1.0 Hz, 1 H), 3.76 (d, J = 3.3 Hz, 1 H), 4.15 (dd, J = 6.3, 3.3 Hz, 1 H), 4.38 (d, J = 11.4 Hz, 1 H), 4.66–4.67 (m, 2 H), 4.94 (d, J = 6.3 Hz, 1 H), 6.01 (d, J = 3.7 Hz, 1 H), 6.99–7.04 (m, 2 H), 7.33–7.39 (m, 5 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 26.1, 26.7, 71.0, 71.9, 81.7, 82.2, 83.5, 105.0, 110.0 (q, J = 8.3 Hz), 112.1, 127.9, 128.4, 128.7, 136.0, 137.5, 140.8, 149.3 (q, J = 6.5 Hz), 152.6 (q, J = 6.4 Hz).

HRMS: m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{21}\text{H}_{21}\text{F}_3\text{O}_5\text{Na}$: 433.1233; found: 433.1235.

3-*O*-Benzyl-5-*C*-4-chlorophenyl-1,2-*O*-isopropylidene- β -L-ido-pentofuranose (5c)

Yellow oil; yield: 0.140 g (0.36 mmol, 72%).

 $[\alpha]_{\text{D}}^{20}$ –26 (*c* 1, CHCl₃).¹H NMR (300 MHz, CDCl₃): δ = 1.31 (s, 3 H), 1.47 (s, 3 H), 2.93 (br s, 1 H), 3.66 (d, *J* = 3.2 Hz, 1 H), 4.24 (dd, *J* = 7.1, 3.3 Hz, 1 H), 4.32 (d, *J* = 11.5 Hz, 1 H), 4.58–4.62 (m, 2 H), 5.02 (d, *J* = 7.0 Hz, 1 H), 6.01 (d, *J* = 3.7 Hz, 1 H), 7.25–7.37 (m, 9 H).¹³C NMR (75 MHz, CDCl₃): δ = 26.1, 26.7, 71.6, 71.7, 81.9, 82.1, 84.2, 105.0, 111.8, 127.6, 128.0, 128.3, 128.3, 128.5, 133.6, 136.7, 138.2.HRMS: *m/z* [M + Na]⁺ calcd for C₂₁H₂₃ClO₅Na: 413.1126; found: 413.1126.**3-*O*-Benzyl-5-*C*-4-bromophenyl-1,2-*O*-isopropylidene- β -L-ido-pentofuranose (5d)**

Yellow oil; yield: 0.160 g (0.37 mmol, 73%).

 $[\alpha]_{\text{D}}^{20}$ –28 (*c* 1, CHCl₃).¹H NMR (300 MHz, CDCl₃): δ = 1.31 (s, 3 H), 1.47 (s, 3 H), 2.92–2.94 (m, 1 H), 3.66 (d, *J* = 3.3 Hz, 1 H), 4.24 (dd, *J* = 7.1, 3.3 Hz, 1 H), 4.32 (d, *J* = 11.5 Hz, 1 H), 4.59–4.62 (m, 2 H), 5.00 (d, *J* = 7.5 Hz, 1 H), 6.00 (d, *J* = 3.7 Hz, 1 H), 7.24–7.43 (m, 9 H).¹³C NMR (75 MHz, CDCl₃): δ = 26.2, 26.8, 71.8, 76.5, 82.0, 82.2, 84.1, 105.1, 112.0, 121.9, 127.7, 128.1, 128.6, 128.7, 131.4, 136.7, 138.7.HRMS: *m/z* [M + Na]⁺ calcd for C₂₁H₂₃BrO₅Na: 457.0621; found: 457.0628.**3-*O*-Benzyl-1,2-*O*-isopropylidene-5-*C*-2-tolyl- β -L-ido-pentofuranose (5e)**

Yellow solid; yield: 0.130 g (0.35 mmol, 70%).

 $[\alpha]_{\text{D}}^{20}$ –23 (*c* 1, CHCl₃).¹H NMR (300 MHz, CDCl₃): δ = 1.32 (s, 3 H), 1.51 (s, 3 H), 2.30 (s, 3 H), 2.75 (br s, 1 H), 3.74 (d, *J* = 3.4 Hz, 1 H), 4.18 (d, *J* = 11.5 Hz, 1 H), 4.46 (d, *J* = 11.5 Hz, 1 H), 4.57–4.59 (m, 2 H), 5.20 (d, *J* = 7.0 Hz, 1 H), 6.04 (d, *J* = 3.7 Hz, 1 H), 7.12–7.30 (m, 8 H), 7.43–7.44 (m, 1 H).¹³C NMR (75 MHz, CDCl₃): δ = 19.2, 26.3, 26.8, 69.1, 71.9, 82.3, 82.6, 83.3, 105.1, 111.8, 126.1, 127.0, 127.7, 127.7, 127.9, 128.4, 130.5, 136.2, 136.7, 137.6.HRMS: *m/z* [M + Na]⁺ calcd for C₂₂H₂₆O₅Na: 393.1672; found: 393.1674.**3-*O*-Benzyl-1,2-*O*-isopropylidene-5-*C*-4-methoxyphenyl- β -L-ido-pentofuranose (5f)**

Yellow oil; yield: 0.147 g (0.38 mmol, 75%).

 $[\alpha]_{\text{D}}^{20}$ –25 (*c* 1, CHCl₃).¹H NMR (300 MHz, CDCl₃): δ = 1.30 (s, 3 H), 1.49 (s, 3 H), 3.60 (d, *J* = 3.3 Hz, 1 H), 3.79 (s, 3 H), 4.27–4.32 (m, 2 H), 4.53 (d, *J* = 11.5 Hz, 1 H), 4.58 (d, *J* = 11.5 Hz, 1 H), 5.00 (d, *J* = 7.6 Hz, 1 H), 6.01 (d, *J* = 3.7 Hz, 1 H), 6.83 (d, *J* = 8.7 Hz, 2 H), 7.25–7.36 (m, 7 H).¹³C NMR (75 MHz, CDCl₃): δ = 26.2, 26.7, 55.2, 71.8, 71.9, 82.1, 82.2, 84.6, 105.1, 111.8, 113.7, 127.6, 127.9, 128.3, 128.5, 131.9, 137.0, 159.0.HRMS: *m/z* [M + Na]⁺ calcd for C₂₂H₂₆O₆Na: 409.1622; found: 409.1625.**3-*O*-Benzyl-1,2-*O*-isopropylidene-5-*C*-4-tolyl- β -L-ido-pentofuranose (5g)**

Yellow oil; yield: 0.126 g (0.34 mmol, 68%).

 $[\alpha]_{\text{D}}^{20}$ –21 (*c* 1, CHCl₃).¹H NMR (300 MHz, CDCl₃): δ = 1.30 (s, 3 H), 1.48 (s, 3 H), 2.34 (s, 3 H), 2.76 (br s, 1 H), 3.63 (d, *J* = 3.2 Hz, 1 H), 4.29 (d, *J* = 3.8Hz, 1 H), 4.30–4.33 (m, 1 H), 4.52–4.59 (m, 2 H), 5.03 (d, *J* = 7.5 Hz, 1 H), 6.01 (d, *J* = 3.7 Hz, 1 H), 7.11–7.36 (m, 9 H).¹³C NMR (75 MHz, CDCl₃): δ = 21.0, 26.2, 26.7, 71.7, 72.1, 82.1, 82.1, 84.5, 105.1, 111.7, 126.9, 127.5, 127.8, 128.4, 128.9, 136.7, 137.0, 137.5.HRMS: *m/z* [M + Na]⁺ calcd for C₂₂H₂₆O₅Na: 393.1672; found: 393.1678.**Aryl-Transfer Reaction to Aldehyde 2; General Procedure**In an atmosphere of argon, 1.5 M Et₂Zn in toluene (2.33 mL, 3.6 mmol, 7.2 equiv) was slowly added to a soln of arylboronic acid (1.2 mmol, 2.4 equiv) in anhyd toluene (2 mL). The mixture was stirred at 60 °C for 1 h and after this period a soln of the aldehyde (0.101 g, 0.5 mmol) was added. The mixture was stirred at 60 °C for 3 h and then extracted with EtOAc (2 × 20 mL). The combined extracts were washed with brine and dried (MgSO₄), and the solvents were evaporated using a rotary evaporator. The residue was purified by flash chromatography (hexane–EtOAc, 7:3).**Methyl 2,3-Isopropylidene-5-*C*-phenyl- α -lyxofuranoside (6a)**

Mixture of diastereomers, data listed for the major diastereomer; yellow oil; yield: 0.110 g (0.39 mmol, 78%).

¹H NMR (300 MHz, CDCl₃): δ = 1.27 (s, 3 H), 1.54 (s, 3 H), 2.92 (br s, 1 H), 3.32 (s, 3 H), 4.01–4.06 (m, 1 H), 4.42–4.45 (m, 1 H), 4.51–4.57 (m, 1 H), 4.95 (s, 1 H), 5.09 (d, *J* = 7.0 Hz, 1 H), 7.29–7.38 (m, 3 H), 7.50–7.54 (m, 2 H).¹³C NMR (75 MHz, CDCl₃): δ = 24.7, 26.1, 54.5, 72.2, 79.8, 83.4, 85.4, 107.2, 112.6, 127.0, 127.9, 128.2, 139.9.HRMS: *m/z* [M + Na]⁺ calcd for C₁₅H₂₀O₅Na: 303.1203; found: 303.1209.**Methyl 5-*C*-4-Chlorophenyl-2,3-isopropylidene- α -lyxofuranoside (6b)**

Mixture of diastereomers, data listed for the major diastereomer; yellow oil; yield: 0.112 g (0.38 mmol, 75%).

¹H NMR (300 MHz, CDCl₃): δ = 1.25 (s, 3 H), 1.55 (s, 3 H), 2.99 (br s, 1 H), 3.31 (s, 3 H), 3.93–3.97 (m, 1 H), 4.42–4.45 (m, 1 H), 4.51–4.55 (m, 1 H), 4.95 (s, 1 H), 5.07 (d, *J* = 7.0 Hz, 1 H), 7.30–7.34 (m, 2 H), 7.44–7.48 (m, 2 H).¹³C NMR (75 MHz, CDCl₃): δ = 24.6, 26.0, 54.7, 71.7, 79.7, 83.2, 85.4, 107.2, 112.7, 128.3, 128.4, 133.6, 138.4.HRMS: *m/z* [M + Na]⁺ calcd for C₁₅H₁₉ClO₅Na: 337.0813; found: 337.0831.**Methyl 2,3-Isopropylidene-5-*C*-4-methoxyphenyl- α -lyxofuranoside (6c)**

Mixture of diastereomers, data listed for the major diastereomer; yellow oil; yield: 0.114 g (0.37 mmol, 74%).

¹H NMR (300 MHz, CDCl₃): δ = 1.26 (s, 3 H), 1.52 (s, 3 H), 2.82 (br s, 1 H), 3.33 (s, 3 H), 3.90 (s, 3 H), 3.98–4.07 (m, 1 H), 4.36–4.42 (m, 1 H), 4.50–4.57 (m, 1 H), 4.95 (s, 1 H), 5.03 (d, *J* = 7.0 Hz, 1 H), 6.86–6.92 (m, 2 H), 7.37–7.45 (m, 2 H).¹³C NMR (75 MHz, CDCl₃): δ = 24.8, 26.1, 54.7, 55.1, 72.0, 79.8, 83.6, 85.4, 107.2, 113.7, 127.7, 128.3, 132.0, 159.3.HRMS: *m/z* [M + Na]⁺ calcd for C₁₆H₂₂O₆Na: 333.1309; found: 333.1332.**Methyl 2,3-Isopropylidene-5-*C*-2-tolyl- α -lyxofuranoside (6d)**

Yellow oil; yield: 0.114 g (0.39 mmol, 78%).

¹H NMR (300 MHz, CDCl₃): δ = 1.23 (s, 3 H), 1.51 (s, 3 H), 2.43 (s, 3 H), 2.80 (br s, 1 H), 3.35 (s, 3 H), 4.24 (dd, *J* = 6.8, 3.3 Hz, 1 H), 4.44 (dd, *J* = 6.0, 3.0 Hz, 1 H), 4.53 (d, *J* = 6.0 Hz, 1 H), 4.98 (s, 1 H), 5.27–5.30 (m, 1 H), 7.15–7.22 (m, 3 H), 7.51–7.55 (m, 1 H).¹³C NMR (75 MHz, CDCl₃): δ = 19.2, 24.6, 26.0, 54.7, 68.7, 80.1, 82.7, 85.4, 107.2, 112.6, 126.0, 127.0, 127.9, 130.5, 136.2, 137.9.

HRMS: m/z $[M + Na]^+$ calcd for $C_{16}H_{22}O_5Na$: 317.1359; found: 317.1380.

Methyl 2,3-Isopropylidene-5-C-4-tolyl- α -lyxofuranoside (6e)

Mixture of diastereomers, data listed for the major diastereomer; yellow oil; yield: 0.117 g (0.40 mmol, 79%).

1H NMR (300 MHz, $CDCl_3$): δ = 1.26 (s, 3 H), 1.54 (s, 3 H), 2.34 (s, 3 H), 2.84 (br s, 1 H), 3.33 (s, 3 H), 4.00–4.04 (m, 1 H), 4.40–4.43 (m, 1 H), 4.50–4.54 (m, 1 H), 4.96 (s, 1 H), 5.05 (d, J = 7.3 Hz, 1 H), 7.14–7.18 (m, 2 H), 7.34–7.41 (m, 2 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 21.1, 24.7, 26.1, 54.6, 72.4, 79.8, 83.5, 85.4, 107.1, 112.6, 127.0, 129.0, 136.9, 137.5.

HRMS: m/z $[M + Na]^+$ calcd for $C_{16}H_{22}O_5Na$: 317.1359; found: 317.1378.

Aryl-Transfer Reaction to Aldehyde 3; General Procedure

In an atmosphere of argon, 1.5 M Et_2Zn in toluene (2.33 mL, 3.6 mmol, 7.2 equiv) was slowly added to a soln of arylboronic acid (1.2 mmol, 2.4 equiv) in anhyd toluene (2 mL). The mixture was stirred at 60 °C for 1 h and after this period a soln of the aldehyde (0.5 mmol, 0.101 g) in toluene (1 mL) was added. The mixture was stirred at 60 °C for 1 h and then extracted with EtOAc (2 $\times$ 20 mL). The combined extracts were washed with brine and dried ($MgSO_4$), and the solvents were evaporated using a rotary evaporator. The residue was purified by flash chromatography (hexane–EtOAc, 7:3).

Methyl 2,3-Isopropylidene-5-C-phenyl- β -ribofuranoside (7a)

Yellow oil; yield: 0.110 g (0.39 mmol, 78%).

$[\alpha]_D^{20}$ –6 (c 1, $CHCl_3$).

1H NMR (300 MHz, $CDCl_3$): δ = 1.31 (s, 3 H), 1.46 (s, 3 H), 3.36 (s, 3 H), 3.83 (d, J = 8.4 Hz, 1 H), 4.66–4.73 (m, 3 H), 4.87 (d, J = 6.2 Hz, 1 H), 4.97 (s, 1 H), 7.26–7.37 (m, 5 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 24.9, 26.4, 56.9, 73.8, 82.0, 85.6, 91.0, 110.3, 112.3, 126.0, 127.5, 128.4, 140.6.

HRMS: m/z $[M + Na]^+$ calcd for $C_{15}H_{20}O_5Na$: 303.1203; found: 303.1230.

Methyl 5-C-4-Chlorophenyl-2,3-isopropylidene- β -ribofuranoside (7b)

Yellow oil; yield: 0.114 g (0.39 mmol, 77%).

$[\alpha]_D^{20}$ –9 (c 1, $CHCl_3$).

1H NMR (300 MHz, $CDCl_3$): δ = 1.32 (s, 3 H), 1.47 (s, 3 H), 3.36 (s, 3 H), 3.90 (d, J = 8.7 Hz, 1 H), 4.62–4.70 (m, 3 H), 4.86 (d, J = 5.8 Hz, 1 H), 4.96 (s, 1 H), 7.31–7.35 (m, 4 H).

^{13}C NMR ($CDCl_3$, 75 MHz): δ = 24.8, 26.3, 55.9, 73.2, 81.8, 85.5, 91.3, 110.3, 112.4, 127.5, 128.5, 133.3, 139.2.

HRMS: m/z $[M + Na]^+$ calcd for $C_{15}H_{19}ClO_5Na$: 337.0813; found: 337.0838.

Methyl 2,3-Isopropylidene-5-C-2-tolyl- β -ribofuranoside (7c)

Yellow oil; yield: 0.111 g (0.38 mmol, 76%).

$[\alpha]_D^{20}$ –9 (c 1, $CHCl_3$).

1H NMR (300 MHz, $CDCl_3$): δ = 1.31 (s, 3 H), 1.47 (s, 3 H), 2.37 (s, 3 H), 3.52 (s, 3 H), 3.78 (d, J = 8.1 Hz, 1 H), 4.60 (d, J = 4.5 Hz, 1 H), 4.67 (d, J = 2.9 Hz, 1 H), 4.79 (d, J = 5.9 Hz, 1 H), 4.88–4.92 (m, 1 H), 5.02 (s, 1 H), 7.17–7.25 (m, 3 H), 7.36–7.39 (m, 1 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 19.4, 24.9, 26.4, 56.3, 71.0, 82.3, 85.6, 90.7, 110.7, 112.3, 126.0, 126.2, 127.6, 130.6, 134.8, 138.3.

HRMS: m/z $[M + Na]^+$ calcd for $C_{16}H_{22}O_5Na$: 317.1359; found: 317.1388.

Methyl 2,3-Isopropylidene-5-C-4-methoxyphenyl- β -ribofuranoside (7d)

Yellow solid; yield: 0.117 g (0.38 mmol, 77%).

$[\alpha]_D^{20}$ –5 (c 1, $CHCl_3$).

1H NMR (300 MHz, $CDCl_3$): δ = 1.30 (s, 3 H), 1.47 (s, 3 H), 3.39 (s, 3 H), 3.64 (d, J = 7.5 Hz, 1 H), 3.80 (s, 3 H), 4.58–4.66 (m, 3 H), 4.80 (d, J = 5.9 Hz, 1 H), 4.98 (s, 1 H), 6.88–6.91 (m, 2 H), 7.26–7.30 (m, 2 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 24.8, 26.4, 55.2, 55.9, 73.6, 81.9, 85.5, 91.6, 110.3, 112.3, 113.9, 127.5, 132.6, 159.0.

HRMS: m/z $[M + Na]^+$ calcd for $C_{16}H_{22}O_6Na$: 333.1309; found: 333.1334.

Methyl 5-C-4-Bromophenyl-2,3-isopropylidene- β -ribofuranoside (7e)

White amorphous solid; yield: 0.129 g (0.36 mmol, 72%).

$[\alpha]_D^{20}$ –6 (c 1, $CHCl_3$).

1H NMR (300 MHz, $CDCl_3$): δ = 1.32 (s, 3 H), 1.48 (s, 3 H), 3.36 (s, 3 H), 3.90 (d, J = 8.6 Hz, 1 H), 4.61–4.69 (m, 3 H), 4.86 (d, J = 5.9 Hz, 1 H), 4.96 (s, 1 H), 7.24 (d, J = 8.0 Hz, 2 H), 7.49 (d, J = 8.0 Hz, 2 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 24.8, 26.3, 56.0, 73.2, 81.8, 86.5, 91.3, 110.3, 112.4, 121.4, 127.8, 131.5, 139.8.

HRMS: m/z $[M + Na]^+$ calcd for $C_{15}H_{19}BrO_5Na$: 381.0308; found: 381.0329.

Aryl-Transfer Reaction to Aldehyde 4; General Procedure

In an atmosphere of argon, 1.5 M Et_2Zn in toluene (2.33 mL, 3.6 mmol, 7.2 equiv) was slowly added to a soln of arylboronic acid (1.2 mmol, 2.4 equiv) in anhyd toluene (2 mL). The mixture was stirred at 60 °C for 1 h and after this period a soln of aldehyde 4 (0.168 g, 0.5 mmol) was added. The mixture was stirred at 60 °C for 3 h and then extracted with EtOAc (2 $\times$ 20 mL). The combined extracts were washed with brine and dried ($MgSO_4$), and the solvents were evaporated using a rotary evaporator. The residue was purified by flash chromatography (hexane–EtOAc, 7:3).

1,2:3,4-Di-*O*-isopropylidene-6-C-phenyl- α -D-galactopyranose (8a)

Mixture of diastereomers; yellow oil; yield: 0.111 g (0.33 mmol, 74%).

1H NMR (300 MHz, $CDCl_3$): δ = 1.26 (s, 3 H), 1.30 (s, 1 H), 1.32 (s, 3 H), 1.36 (s, 1 H), 1.48 (s, 1 H), 1.51 (s, 3 H), 1.52 (s, 3 H), 1.54 (s, 1 H), 3.0 (s, 1 H), 3.30 (d, J = 6.2 Hz, 0.3 H), 3.73–3.77 (m, 1 H), 3.82–3.85 (m, 1 H), 3.91–3.94 (m, 0.3 H), 4.28–4.31 (m, 0.3 H), 4.36–4.39 (m, 1 H), 4.44–4.47 (m, 0.3 H), 4.88–4.96 (m, 1.3 H), 5.53 (d, J = 5.0 Hz, 0.3 H), 5.63 (d, J = 4.8 Hz, 1 H), 7.29–7.5 (m, 6.5 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 24.1, 24.3, 24.8, 24.9, 25.9, 26.0, 69.8, 70.4, 70.6, 70.7, 70.8, 70.9, 71.1, 128.3, 128.4, 139.2, 141.0.

HRMS: m/z $[M + Na]^+$ calcd for $C_{18}H_{24}O_6Na$: 359.1483; found: 359.1510.

1,2:3,4-Di-*O*-isopropylidene-6-C-4-tolyl- α -D-galactopyranose (8b)

Mixture of diastereomers; yellow oil; yield: 0.126 g (0.36 mmol, 72%).

1H NMR (300 MHz, $CDCl_3$): δ = 1.25 (s, 1 H), 1.27 (s, 1 H), 1.29 (s, 3 H), 1.32 (s, 1 H), 1.35 (s, 3 H), 1.49 (s, 3 H), 1.52 (s, 1 H), 1.53 (s, 3 H), 2.34 (s, 4 H), 3.0 (s, 0.4 H), 3.18 (d, J = 6.15 Hz, 1 H), 3.72–3.75 (m, 0.4 H), 3.82–3.85 (m, 0.4 H), 3.90–3.93 (m, 1 H), 4.28–4.30 (m, 1.4 H), 4.37–4.40 (m, 1 H), 4.42–4.46 (m, 0.4 H), 4.55–4.58 (m, 1 H), 4.85–4.92 (m, 1.4 H), 5.55 (d, J = 5.0 Hz, 1 H), 5.60 (d, J = 4.8 Hz, 0.4 H), 7.10–7.32 (m, 5.6 H).

^{13}C NMR (75 MHz, $CDCl_3$): δ = 21.1, 24.2, 24.3, 24.8, 24.9, 25.9, 26.1, 69.7, 70.5, 70.8, 70.9, 71.0, 71.1, 72.4, 72.7, 73.8, 96.5, 96.7, 108.5, 108.8, 109.2, 109.5, 126.4, 127.2, 128.9, 129.0, 137.3, 138.3.

HRMS: m/z $[M + Na]^+$ calcd for $C_{19}H_{26}O_6Na$: 373.1622; found: 373.1639.

1,2:3,4-Di-*O*-isopropylidene-6-*C*-4-methoxyphenyl- α -D-galactopyranose (8c)

Mixture of diastereomers; yellow oil; yield: 0.119 g (0.325 mmol, 65%).

^1H NMR (300 MHz, CDCl_3): δ = 1.27 (s, 3 H), 1.30 (s, 0.6 H), 1.32 (s, 3 H), 1.36 (s, 0.6 H), 1.49 (s, 0.6 H), 1.50 (s, 3 H), 1.53 (s, 3 H), 1.56 (s, 0.6 H), 3.0 (s, 1 H), 3.07 (d, J = 5.8 Hz, 0.6 H), 3.71–3.74 (m, 1 H), 3.80 (s, 1.8 H), 3.81 (s, 3 H), 3.81–3.84 (m, 1 H), 4.28–4.31 (m, 1.6 H), 4.39–4.47 (m, 1.6 H), 4.57–4.60 (m, 0.6 H), 4.83–4.90 (m, 1.6 H), 5.54 (d, J = 5.0 Hz, 0.6 H), 5.60 (d, J = 4.9 Hz, 1 H), 6.87–6.91 (m, 3.2 H), 7.33–7.42 (m, 3.2 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 69.8, 70.4, 70.7, 70.8, 70.9, 71.1, 72.2, 72.7, 73.3, 96.5, 96.6, 108.5, 108.7, 109.2, 109.4, 113.6, 113.7, 127.7, 128.4, 131.3, 133.5, 159.1, 159.3.

HRMS: m/z [$\text{M} + \text{Na}$] $^+$ calcd for $\text{C}_{19}\text{H}_{26}\text{O}_7\text{Na}$: 389.1571; found: 389.1582.

3-*O*-Benzyl-1,2-*O*-isopropylidene-5-*C*-phenylethynyl- β -L-idopentofuranose (9)

In an atmosphere of argon, 1.5 M Et_2Zn in toluene (1.33 mL, 2 mmol, 4 equiv) was slowly added to a soln of phenylacetylene (4.0 equiv, 2.0 mmol) in anhyd toluene (2 mL). The mixture was stirred for 30 min and a soln of the aldehyde (0.5 mmol, 0.139 g) in anhyd toluene (2 mL) was slowly added. The mixture was stirred at r.t. for 15 h and then extracted with EtOAc (3×20 mL). The combined extracts were washed with brine and dried (MgSO_4), and the solvents were evaporated on a rotary evaporator. The residue was purified by flash chromatography (hexane– EtOAc , 7:3), to give the product as a mixture of diastereomers (data listed for the major diastereomer); yellow oil; yield: 0.122 g (0.32 mmol, 64%).

^1H NMR (300 MHz, CDCl_3): δ = 1.32 (s, 3 H), 1.51 (s, 3 H), 2.90 (br s, 1 H), 3.69 (d, J = 8.7 Hz, 1 H), 4.16 (d, J = 3.3 Hz, 1 H), 4.38–4.44 (m, 2 H), 4.55–4.59 (m, 1 H), 4.98 (d, J = 8.4 Hz, 1 H), 6.00 (d, J = 3.7 Hz, 1 H), 7.24–7.42 (m, 10 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 26.4, 26.9, 62.3, 72.8, 81.8, 82.3, 83.9, 86.3, 87.5, 105.6, 112.2, 122.2, 128.0, 128.1, 128.3, 128.5, 128.6, 131.8, 137.1.

Conversion of 5a–e into 11a–e; General Procedure

A soln of 5a–e (1 mmol) in 70% aq AcOH (10 mL) was stirred for 5 h under reflux. After this time, the mixture was cooled to r.t. and concentrated by co-distillation with toluene. The residue was filtered through a plug of silica (EtOAc), the solvent removed under vacuum and the products 10a–e were used directly in the next step. To a soln of the obtained product in anhyd DMF (2 mL) were added Meldrum's acid (0.288 g, 2.0 mmol) and anhyd Et_3N (0.280 mL, 2.0 mmol). The mixture was stirred for 72 h at 46 °C and the solvents were evaporated using a rotary evaporator. The residue was purified by flash chromatography (Et_2O).

3,6-Anhydro-5-*O*-benzyl-2-deoxy-7-*C*-phenyl-L-glycero-D-ido-heptono-1,4-lactone (11a)

White amorphous solid; yield: 0.086 g (0.255 mmol, 51%).

$[\alpha]_{\text{D}}^{20} +38$ (c 0.5, CHCl_3).

^1H NMR (300 MHz, CDCl_3): δ = 2.72–2.73 (m, 3 H), 3.91 (d, J = 3.9 Hz, 1 H), 4.23 (dd, J = 6.6, 3.9 Hz, 1 H), 4.46 (d, J = 11.7 Hz, 1 H), 4.55 (d, J = 11.7 Hz, 1 H), 4.93 (d, J = 4.2 Hz, 1 H), 5.03 (d, J = 6.9 Hz, 1 H), 5.07–5.11 (m, 1 H), 7.31–7.39 (m, 10 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 36.0, 72.8, 81.8, 84.6, 85.1, 126.8, 127.8, 128.2, 128.3, 128.4, 128.6, 136.5, 139.5, 175.2.

3,6-Anhydro-5-*O*-benzyl-2-deoxy-7-*C*-3,4,5-trifluorophenyl-L-glycero-D-ido-heptono-1,4-lactone (11b)

Yellow oil; yield: 0.110 g (0.28 mmol, 56%).

$[\alpha]_{\text{D}}^{20} +55$ (c 0.5, CHCl_3).

^1H NMR (300 MHz, CDCl_3): δ = 2.73 (m, 3 H), 4.14 (m, 2 H), 4.52 (d, J = 11.3 Hz, 1 H), 4.70 (d, J = 11.3 Hz, 1 H), 4.92 (d, J = 4.6 Hz,

1 H), 4.97 (d, J = 4.6 Hz, 1 H), 5.07–5.11 (m, 1 H), 6.92–7.01 (m, 2 H), 7.34–7.41 (m, 5 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 35.9, 71.4, 72.9, 82.0, 83.8, 85.0, 110.7 (q, J = 8.4 Hz), 111.0, 127.8, 128.0, 128.6, 128.7, 128.8, 135.9, 149.3 (q, J = 6.5 Hz), 152.7 (q, J = 6.5 Hz), 174.8.

3,6-Anhydro-5-*O*-benzyl-7-*C*-4-chlorophenyl-2-deoxy-L-glycero-D-ido-heptono-1,4-lactone (11c)

Colorless oil; yield: 0.099 g (0.265 mmol, 53%).

$[\alpha]_{\text{D}}^{20} +68$ (c 0.5, CHCl_3).

^1H NMR (300 MHz, CDCl_3): δ = 2.71–2.73 (m, 2 H), 3.96 (d, J = 4.2 Hz, 1 H), 4.18 (dd, J = 6.3, 4.2 Hz, 1 H), 4.45 (d, J = 11.7 Hz, 1 H), 4.60 (d, J = 11.7 Hz, 1 H), 4.93 (d, J = 3.9 Hz, 1 H), 4.99 (d, J = 6.3 Hz, 1 H), 5.07–5.11 (m, 1 H), 7.29–7.40 (m, 9 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 36.0, 65.8, 72.2, 81.9, 84.4, 85.0, 127.8, 128.2, 128.4, 128.6, 128.7, 133.9, 136.3, 138.0, 175.0.

3,6-Anhydro-5-*O*-benzyl-7-*C*-4-bromophenyl-2-deoxy-L-glycero-D-ido-heptono-1,4-lactone (11d)

Colorless oil; yield: 0.114 g (0.275 mmol, 55%).

$[\alpha]_{\text{D}}^{20} +44$ (c 0.5, CHCl_3).

^1H NMR (300 MHz, CDCl_3): δ = 2.71–2.73 (m, 2 H), 3.97 (d, J = 4.2 Hz, 1 H), 4.19 (dd, J = 6.0, 4.2 Hz, 1 H), 4.45 (d, J = 11.7 Hz, 1 H), 4.61 (d, J = 11.7 Hz, 1 H), 4.93 (d, J = 4.5 Hz, 1 H), 4.98 (d, J = 6.0 Hz, 1 H), 5.07–5.11 (m, 1 H), 7.22–7.47 (m, 9 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 36.0, 72.2, 72.8, 76.6, 81.9, 84.4, 85.0, 122.1, 127.8, 128.5, 128.7, 131.5, 136.3, 138.5, 175.0.

3,6-Anhydro-5-*O*-benzyl-2-deoxy-7-*C*-2-tolylphenyl-L-glycero-D-ido-heptono-1,4-lactone (11e)

Yellow oil; yield: 0.089 g (0.252 mmol, 51%).

$[\alpha]_{\text{D}}^{20} +53$ (c 0.5, CHCl_3).

^1H NMR (300 MHz, CDCl_3): δ = 2.27–2.29 (m, 4 H), 2.73–2.74 (m, 2 H), 4.04–4.07 (m, 1 H), 4.36 (d, J = 11.7 Hz, 1 H), 4.49 (d, J = 11.7 Hz, 1 H), 4.54 (dd, J = 6.0, 4.5 Hz, 1 H), 4.91 (d, J = 4.6 Hz, 1 H), 5.11–5.14 (m, 1 H), 5.20 (d, J = 5.7 Hz, 1 H), 7.18–7.36 (m, 9 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 19.2, 36.0, 69.3, 72.8, 82.2, 83.4, 85.5, 126.2, 126.7, 127.9, 128.0, 128.3, 128.6, 130.5, 136.4, 136.6, 137.6, 175.0.

Hydrogenolysis of 11a–e and Synthesis of (+)-7-*epi*-Goniofufurone and Analogues; General Procedure

A soln of 11 (0.5 mmol) in MeOH (2 mL) was hydrogenated over 10% Pd/C for 120 h at r.t. The mixture was filtered and the catalyst washed with EtOAc . The combined organic solns were evaporated in a rotary evaporator and the residue was purified by flash chromatography (CH_2Cl_2 –MeOH, 98:2–95:5).

(+)-7-*epi*-Goniofufurone (12a)

Colorless amorphous solid; yield: 0.066 g (0.265 mmol, 73%).

$[\alpha]_{\text{D}}^{20} +102$ (c 0.3, EtOH).

^1H NMR (300 MHz, CD_3OD): δ = 2.15 (s, 1 H), 2.60 (d, J = 18.6 Hz, 1 H), 2.86 (dd, J = 18.6, 6.3 Hz, 1 H), 3.78 (d, J = 2.6 Hz, 1 H), 3.97 (dd, J = 7.9, 2.9 Hz, 1 H), 4.81 (d, J = 4.2 Hz, 1 H), 4.92 (d, J = 8.4 Hz, 1 H), 5.02–5.06 (m, 1 H), 7.28–7.36 (m, 3 H), 7.44–7.47 (m, 2 H).

^{13}C NMR (75 MHz, CD_3OD): δ = 36.8, 73.9, 74.6, 78.6, 86.3, 89.7, 128.2, 129.0, 129.3, 142.4, 178.2.

HRMS: m/z [$\text{M} + \text{Na}$] $^+$ calcd for $\text{C}_{13}\text{H}_{14}\text{O}_5\text{Na}$ + MeOH: 305.0996; found: 305.0993.

3,6-Anhydro-2-deoxy-7-*C*-3,4,5-trifluorophenyl-L-glycero-D-ido-heptono-1,4-lactone (12b)

White solid; yield: 0.109 g (0.36 mmol, 72%).

$[\alpha]_{\text{D}}^{20} +57$ (c 0.3, EtOH).

^1H NMR (300 MHz, CD_3OD): δ = 2.72–2.76 (m, 2 H), 3.47 (d, J = 5.0 Hz, 1 H), 3.72 (s, 1 H), 4.08–4.11 (m, 1 H), 4.44 (d, J = 3.2 Hz, 1 H), 4.90 (d, J = 3.8 Hz, 1 H), 5.02 (d, J = 4.2 Hz, 1 H), 5.09–5.13 (m, 1 H), 7.07–7.12 (m, 2 H).

^{13}C NMR (75 MHz, CD_3OD): δ = 36.8, 72.5, 74.8, 78.8, 86.0, 89.8, 112.2 (q, J = 8.3 Hz), 112.5, 139.8, 139.9, 150.6 (q, J = 6.5 Hz), 153.8 (q, J = 6.4 Hz), 178.0.

HRMS: m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{11}\text{F}_3\text{O}_5\text{Na} + \text{MeOH}$: 359.0713; found: 359.0720.

3,6-Anhydro-7-C-4-chlorophenyl-2-deoxy-L-glycero-D-ido-heptono-1,4-lactone (12c)

Colorless oil; yield: 0.079 g (0.28 mmol, 66%).

$[\alpha]_{\text{D}}^{20} +25$ (c 0.3, EtOH).

^1H NMR (300 MHz, CD_3OD): δ = 2.72–2.75 (m, 2 H), 3.47 (s, 1 H), 3.48 (s, 1 H), 4.16–4.18 (m, 1 H), 4.41–4.42 (m, 1 H), 4.80 (d, J = 3.2 Hz, 1 H), 5.01 (d, J = 3.0 Hz, 1 H), 5.09–5.12 (m, 1 H), 7.30–7.36 (m, 4 H).

^{13}C NMR (75 MHz, CD_3OD): δ = 36.8, 73.9, 74.7, 77.8, 83.4, 89.8, 127.3, 129.4, 130.3, 139.8, 173.9.

HRMS: m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{13}\text{ClO}_5\text{Na} + \text{MeOH}$: 339.0606; found: 339.0605.

3,6-Anhydro-5-O-benzyl-7-C-4-bromophenyl-2-deoxy-L-glycero-D-ido-heptono-1,4-lactone (12d)

White solid; yield: 0.111 g (0.340 mmol, 68%).

$[\alpha]_{\text{D}}^{20} +87$ (c 0.3, EtOH).

^1H NMR (300 MHz, CD_3OD): δ = 2.14 (s, 1 H), 2.60 (d, J = 18.0 Hz, 1 H), 2.86 (dd, J = 18.6, 6.3 Hz, 1 H), 3.66 (s, 1 H), 3.99 (dd, J = 9.3, 2.9 Hz, 1 H), 4.82 (d, J = 3.7 Hz, 1 H), 4.91 (d, J = 7.9 Hz, 1 H), 5.02–5.06 (m, 1 H), 7.30–7.47 (m, 4 H).

^{13}C NMR (75 MHz, CD_3OD): δ = 36.8, 73.9, 74.6, 78.6, 86.3, 89.7, 128.2, 129.0, 129.3, 142.4, 178.2.

HRMS: m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{13}\text{BrO}_5\text{Na} + \text{MeOH}$: 383.0101; found: 383.0101.

3,6-Anhydro-5-O-benzyl-2-deoxy-7-C-2-tolylphenyl-L-glycero-D-ido-heptono-1,4-lactone (12e)

Colorless amorphous solid; yield: 0.075 g (0.285 mmol, 57%).

$[\alpha]_{\text{D}}^{20} +43$ (c 0.3, EtOH).

^1H NMR (300 MHz, CD_3OD): δ = 2.15 (s, 1 H), 2.45 (s, 3 H), 2.60 (d, J = 18.6 Hz, 1 H), 2.88 (dd, J = 18.6 Hz, 6.3 Hz, 1 H), 3.85 (d, J = 2.8 Hz, 1 H), 4.26 (dd, J = 7.6 Hz, 3.1 Hz, 1 H), 4.81 (d, J = 3.6 Hz, 1 H), 5.03–5.07 (m, 1 H), 7.14–7.21 (m, 3 H), 7.39–7.42 (m, 1 H).

^{13}C NMR (75 MHz, CD_3OD): δ = 19.5, 36.9, 70.1, 75.1, 78.5, 85.9, 89.9, 128.1, 128.9, 129.0, 129.2, 131.5, 137.4, 178.2.

HRMS: m/z $[\text{M} + \text{Na}]^+$ calcd for $\text{C}_{13}\text{H}_{14}\text{O}_5\text{Na} + \text{MeOH}$: 319.1152; found: 319.1151.

Acknowledgment

We are grateful to FAPESP, CNPq, and FAPERGS for financial support. CAPES is acknowledged for Ph.D. fellowships to A.D.W. and M.S., and Sandwich Ph.D. fellowship to A.D.W.; D.S.L. is the recipient of a CNPq research fellowship. We also thank Dr. Angélica V. Moro for helpful discussions and proofreading the manuscript.

Supporting Information for this article is available online at <http://www.thieme-connect.com/ejournals/toc/synthesis>.

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